Epidemiology of and Treatments for Ligneous Conjunctivitis: A Literature Review Amid a Surging Demand for Plasminogen Eye Drops Prepared by Héma-Québec

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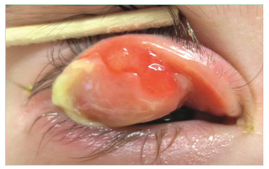
Background

- Ligneous conjunctivitis (LC) is the most common manifestation of congenital type 1 plasminogen (Plg) deficiency (cPD), an autosomal recessive disorder caused by mutations in the gene that encodes Plg (**Figure 1**).[1,2]
  - The epidemiology of cPD (and, by extension, LC) is not well understood because of the rarity of this condition, which might affect <1.6 individuals per 1,000,000 population.[3]

Figure 1. A pediatric patient with ligneous conjunctivitis



- The management of LC is not standardized, and most treatment options offer modest or no improvements.[3]
  - Plg eye drops might be a promising treatment for LC, but only a limited number of patients described in the literature have received this treatment.
- Héma-Québec (HQ) is the only provider of Plg eye drops in Québec.
  - Until recently, only one patient with LC had been initiated on HQ-prepared Plg eye drops.
- However, three more patients with LC have recently initiated this treatment.
- Whether this surging demand may grow further is unclear, in part because the characteristics of patients with LC and their treatments are unclear.

Objective

- **To synthesize data on published cases of LC, with a view to better understand the epidemiology of and treatments for this rare condition.**

Methods

Literature search

- On May 5, 2021, we searched MEDLINE records using the following terms: (Ligneous conjunctivitis [Title/Abstract] OR Conjunctivitis lignosa [Title/Abstract]) AND (case* [Title/Abstract] OR child [Title/Abstract] OR boy* [Title/Abstract] OR girl* [Title/Abstract] OR infant* [Title/Abstract] OR patient* [Title/Abstract]) AND ("1997/01/01"[Date - Entry] : "2021/05/06"[Date - Entry]).
- We focused on articles published in 1997 or later owing to concerns related to diagnostic accuracy prior to the identification of cPD as the predominant cause of LC.
- All patients with LC were included in the review; no other inclusion or exclusion criteria applied.

Data extraction & statistical analysis

- The following variables were extracted: sex, age at symptom onset, duration of symptoms, and Plg activity (measured as a percentage of normal activity).
- Descriptive summary statistics were reported, and the characteristics of male and female patients were compared with the Wilcoxon-Mann-Whitney test.

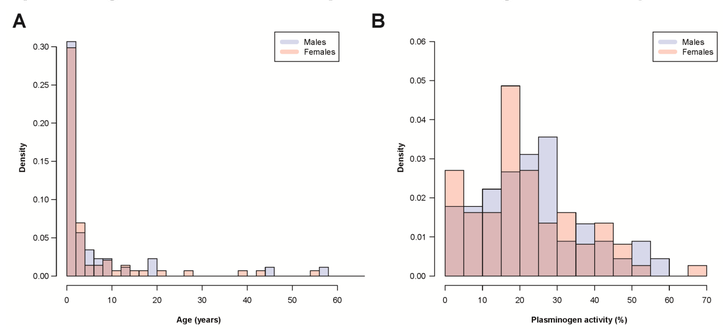
Results

- In total, 78 articles reported data on **210 patients**.

Epidemiology

- Overall, 130 (63.1%) patients with LC were females (among the 206 with available data), a proportion consistent with that reported in an international registry of patients with cPD (62.5%).[4]
- The median age of symptom onset was 1.4 years and was similar in males and females (p=0.9443; **Figure 2A**).
  - Overall, 99 (83.9%) of patients with LC experienced their first symptoms before <10 years.

Figure 2. Distribution of (A) age at symptom onset (N=118 with available data) and (B) Plg activity (N=129 patients with available data) among male and female patients with ligneous conjunctivitis

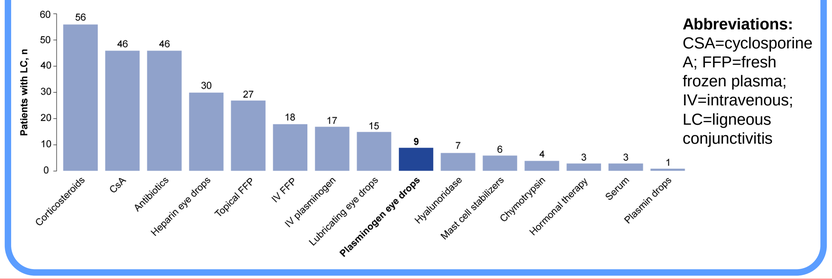


- The median Plg activity, measured as a percentage of normal activity, was 21.5% and was similar in males and females (p=0.2559; **Figure 2B**).
  - Overall, only 32 (24.8%) patients had a Plg activity >30% of normal.

Treatment patterns

- At the patient level, local or systemic corticosteroids (N=56 [47.9%]), antibiotics (N=46 [39.3%]), cyclosporine A (N=46 [39.3%]), heparin (N=30 [25.6%]), and fresh frozen plasma eye drops (N=27 [23.1%]) were the most common non-surgical interventions (**Figure 3**).
- Only nine patients (7.7%) received Plg eye drops (**Figure 3**).
  - Seven articles explicitly mentioned the lack of a commercial preparation as a reason for not initiating this treatment.

Figure 3. Use of non-surgical treatments for ligneous conjunctivitis



Limitations

- In two patients, LC seemingly ensued from a chemical injury (one with hydrochloric acid, and one with tranexamic acid) and was therefore not due to cPD. Whether these cases can be considered a subtype of LC or a separate disease entity is currently unclear.
- In 33 (15.7%) patients, Plg activity — or any other confirmation of cPD — was not reported.
  - However, the vast majority of these patients had available *PLG* sequencing data or a family history that strongly pointed to cPD as the cause of LC.
- Data on treatment effectiveness could not be analyzed, because of significant heterogeneity in the reporting of patient outcomes.
  - Nonetheless, based on qualitative effectiveness measurements, Plg eye drops appeared more efficacious than all other treatments.
- As any other literature review, this study is subject to publication biases.
  - For example, 82 (39.0%) of the patients were identified in case reports or small case series of 2-3 patients. The patients included in these small reports may show features (e.g., rare comorbidities) that are poorly representative of the overall population of patients with LC.

Conclusions

- To the best of our knowledge, no prior reviews have synthesized data on so many documented cases of LC.
- The first symptoms of LC typically manifest during early childhood, so that patients may need lifelong treatment.
- The treatment patterns for LC were highly heterogeneous, and few patients in our review received Plg eye drops, at least in part due to the lack of a commercial preparation.
- We conclude that the demand for Plg eye drops is poised to grow further as healthcare providers become aware of the apparent effectiveness of this treatment. Nonetheless, more studies are needed to assess the efficacy of Plg eye drops.

References

1. Mingers AM, Heimburger N, Zeitler P, Kreth HW, & Schuster V. *Semin. Thromb. Hemost.* **23**, 259–269 (1997).
2. Mingers AM, Philapitsch A, Schwarz HP, Zeitler P, & Kreth HW. *Semin. Thromb. Hemost.* **24**, 605–612 (1998).
3. Schuster V & Seregard S. *Surv. Ophthalmol.* **48**, 369–388 (2003).
4. Shapiro AD *et al. Haematologica* **105**, 554–561 (2020).

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Conflicts of interest: The authors have nothing to declare.